

ARPITA VERMA

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EDUCATION

Amity University Madhya Pradesh

Bachelor of Technology in Computer Science Engineering

Aug 2023 – May 2027

Specialization - Artificial Intelligence and Machine Learning

CGPA: 9.56/10.00

Skomiya Academy, Kanpur (CBSE)

2021-22

Intermediate (PCM)

Percentage: 92 %

Skomiya Academy, Kanpur (CBSE)

2019-20

High School

Percentage: 93.2 %

EXPERIENCE

AI/ML Intern | Defense-Oriented UAV Identification System

July 2025 - August 2025

- Completed an on-site internship at a government defense facility, working on a UAV-based identification system under real-world operational constraints
- Developed an EO/IR-based machine learning Identification Friend or Foe (IFF) system for UAV applications
- Implemented CUDA-based optimizations to improve real-time inference performance in constrained environments
- Managed end-to-end pipeline including data collection, preprocessing, model training, and evaluation
- Deployed the system in offline environments ensuring reliability and robustness
- Analyzed model performance using Python and Excel for evaluation and optimization.

PROJECTS

Military Vehicle Detection System (Computer Vision) Framework

Python, YOLO, OpenCV, PyTorch, CUDA, Qt

- Trained a YOLO-based object detection model on a custom annotated dataset for military vehicles.
- Performed dataset annotation and preprocessing to improve detection accuracy.
- Built a complete offline pipeline for training and real-time inference.
- Developed a Qt-based GUI for deployment and visualization.
- Optimized performance using CUDA for efficient detection.

Traffic Violation Detection System (Computer Vision – In Progress)

- Developing a system to detect traffic violations such as helmet absence and triple riding
- Exploring object detection models (YOLO / RT-DETR) for real-time performance and accuracy trade-offs
- Working on dataset selection and preprocessing, including potential custom annotation
- Designing logic to count riders and detect violations in video streams
- Planning model training, evaluation, and integration with real-time video streams
- Tools: Python, OpenCV, PyTorch

Velvet Wallet – House Budget Planner App (Android)

Kotlin, Android Studio, Room Database

- Developed an Android application for managing personal and household expenses
- Implemented income/expense tracking, budget monitoring, and real-time balance updates
- Designed a user-friendly interface for financial planning
- Used Room Database for efficient and structured local data storage

TECHNICAL SKILLS

Languages: C/C++, Python, Java

AI/ML & Libraries: PyTorch, TensorFlow, OpenCV, Scikit-learn

Tools & Technologies: CUDA, Git, GitHub

Web: HTML, CSS, JavaScript

Databases: MySQL

Data Analysis: Pandas, NumPy

Core Subjects: Data Structures & Algorithms, OS, OOP, DBMS

ACHIEVEMENTS

- Successfully completed an on-site AI/ML internship at a government defense facility, awarded an official internship certificate.
- Qualified GATE 2026 – Computer Science
- Selected for Smart India Hackathon (SIH) 2025 and advanced to next round
- Participated in Yukti Hackathon 2024 (24-hour challenge)
- Built an AI-powered system in Novo Nordisk GBS Hackathon 2025 for automated data visualization

LINKS

GitHub: <https://github.com/arpita080404>